

MINERALOCORTICOIDS

- **Corticosteroids that can act on minerals (electrolytes)**, particularly Na^+ & K^+ .
- **Mineralocorticoids** →
 - a. Aldosterone
 - b. 11-Deoxycorticosterone

SOURCE OF SECRETION

- **Secreted by:** Zona glomerulosa of adrenal cortex.

CHEMISTRY

- C_{21} steroids (21 carbon atoms).
- **Half-life:** 20 minutes.

PLASMA LEVEL & DAILY OUTPUT

- **Plasma level :—**
 - a. Aldosterone: $0.006 \mu\text{g/dL}$
 - b. 11-Deoxycorticosterone: $0.006 \mu\text{g/dL}$
- **Daily output :—**
 - a. Aldosterone: $0.15 \mu\text{g}$
 - b. 11-Deoxycorticosterone: $0.2 \mu\text{g}$

FUNCTIONS OF MINERALOCORTICOIDS

- **90% mineralocorticoid activity** provided by **aldosterone**.
- **Life-saving hormone:** Aldosterone maintains osmolarity and volume of ECF.

Aldosterone has 3 important functions :—

- a. Reabsorption of Na^+ from renal tubules.
- b. Excretion of K^+ through renal tubules.
- c. Secretion of H^+ ion into renal tubules.

ACTIONS OF ALDOSTERONE

1. On Sodium Ions

- **Aldosterone acts on** DCT and collecting duct.
- **↑se reabsorption** of Na^+ .
- **Hyposecretion of aldosterone:**
 - ↔ Loss of Na^+ through urine ↑es (**hyponatriuria**) up to about 20 g/day.

2. On ECF

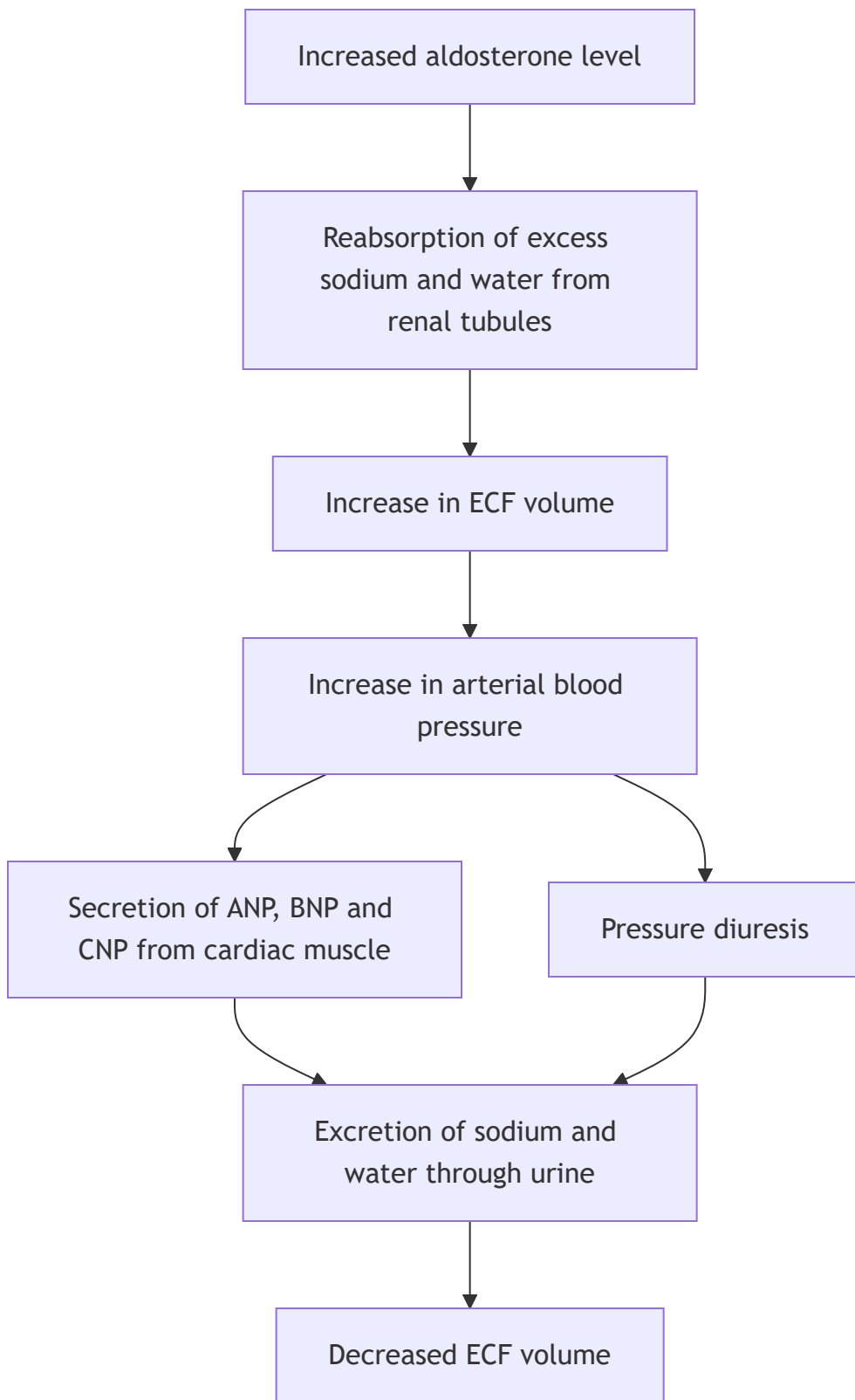
- When Na^+ are reabsorbed from renal tubules, **simultaneously water is also reabsorbed:**
 - ↓
 - So, net result is the **↑se in ECF volume**.
- **Mild ↑se in concentration of Na^+ in blood (mild hypernatremia):**
 - ↓
 - Induce **thirst** → again ↑ses ECF volume & blood volume.

3. On Blood Pressure

- ↑se in ECF volume and blood volume → **leads to ↑se in BP**.

Aldosterone Escape / Escape Phenomenon

- **Aldosterone escape refers to** escape of kidney from salt-retaining effects of excess administration or secretion of aldosterone:
 - ↓
 - In case of **hyperaldosteronism**.
- **Because of aldosterone escape, edema does not occur.**



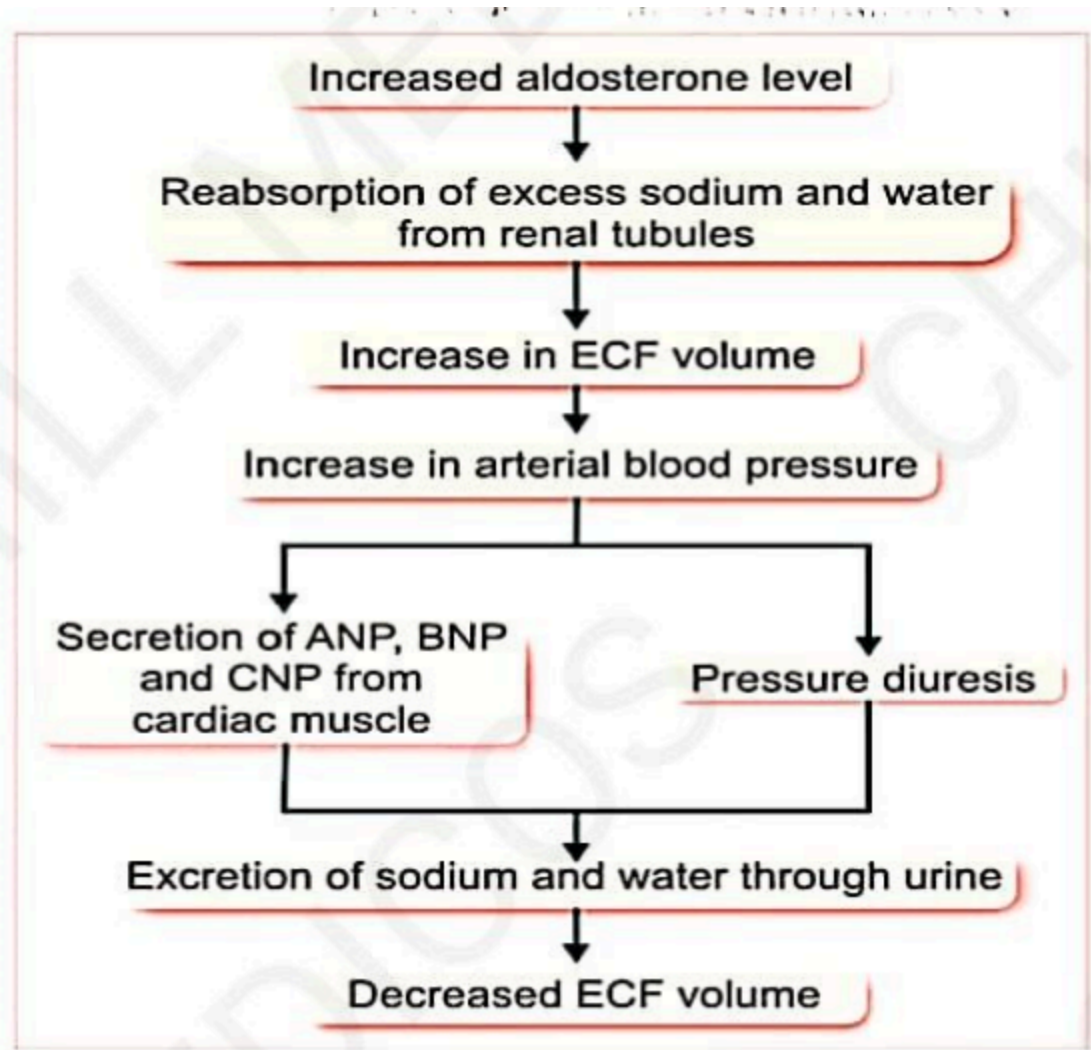


FIGURE 70.3: Aldosterone escape

4. On K^+

- Aldosterone ↑ses K^+ excretion through renal tubules.
- When aldosterone deficient:
 ⇨ K^+ concentration in ECF ↑ses leading to **hyperkalemia** → leads to development of **arrhythmia**.
- When aldosterone secretion ↑ses:
 ⇨ Leads to **hypokalemia** and **muscular weakness**.

5. On H^+ Concentration

- While ↑sing Na^+ reabsorption from renal tubules, **aldosterone causes tubular secretion of hydrogen ions**.
- Aldosterone maintains acid–base balance in body:
- Hypersecretion → Causes **alkalosis**

- Hyposecretion → Causes **acidosis**

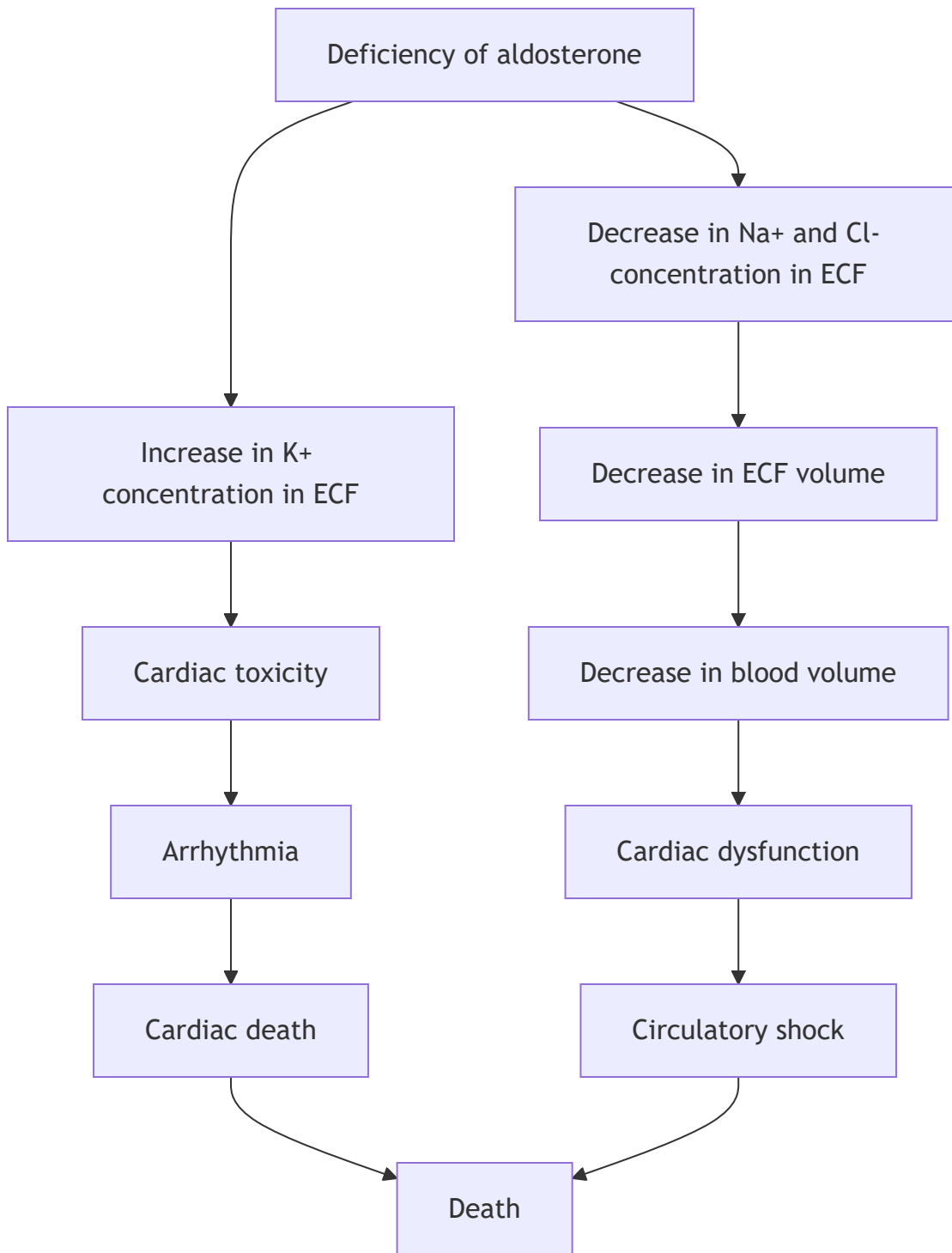
6. On Sweat Glands & Salivary Glands

- Na^+ reabsorbed from sweat glands under influence of aldosterone, **thus loss of Na^+ from body is prevented.**
- **Same effect on saliva also.**

7. On Intestine

- **Aldosterone ↑ses Na^+ absorption from intestine**, especially from colon:
- ↓
- Prevents loss of Na^+ through feces.
- **Aldosterone deficiency leads to diarrhea** (*with loss of Na^+ & H_2O*).

IMPORTANCE OF ALDOSTERONE



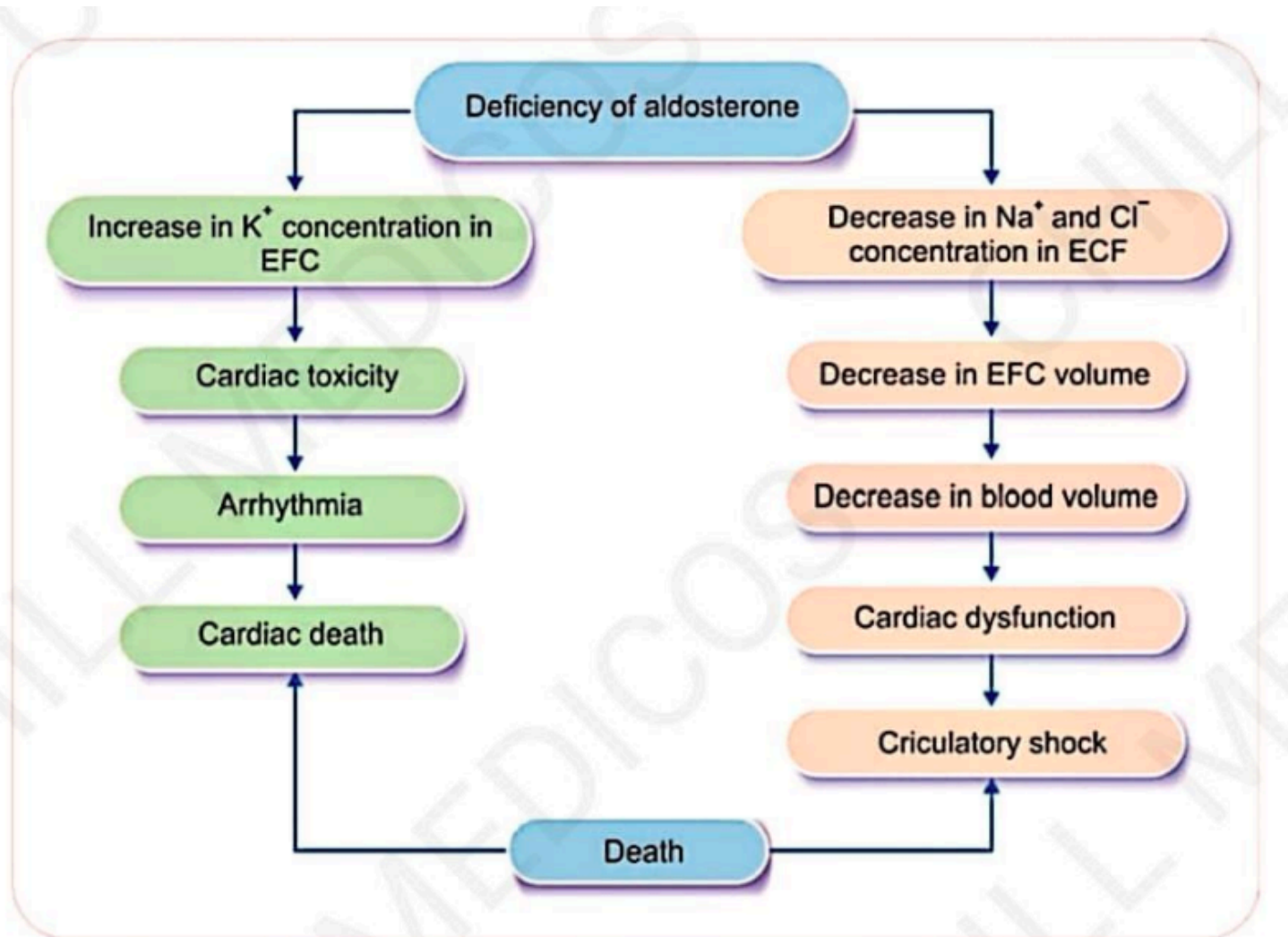
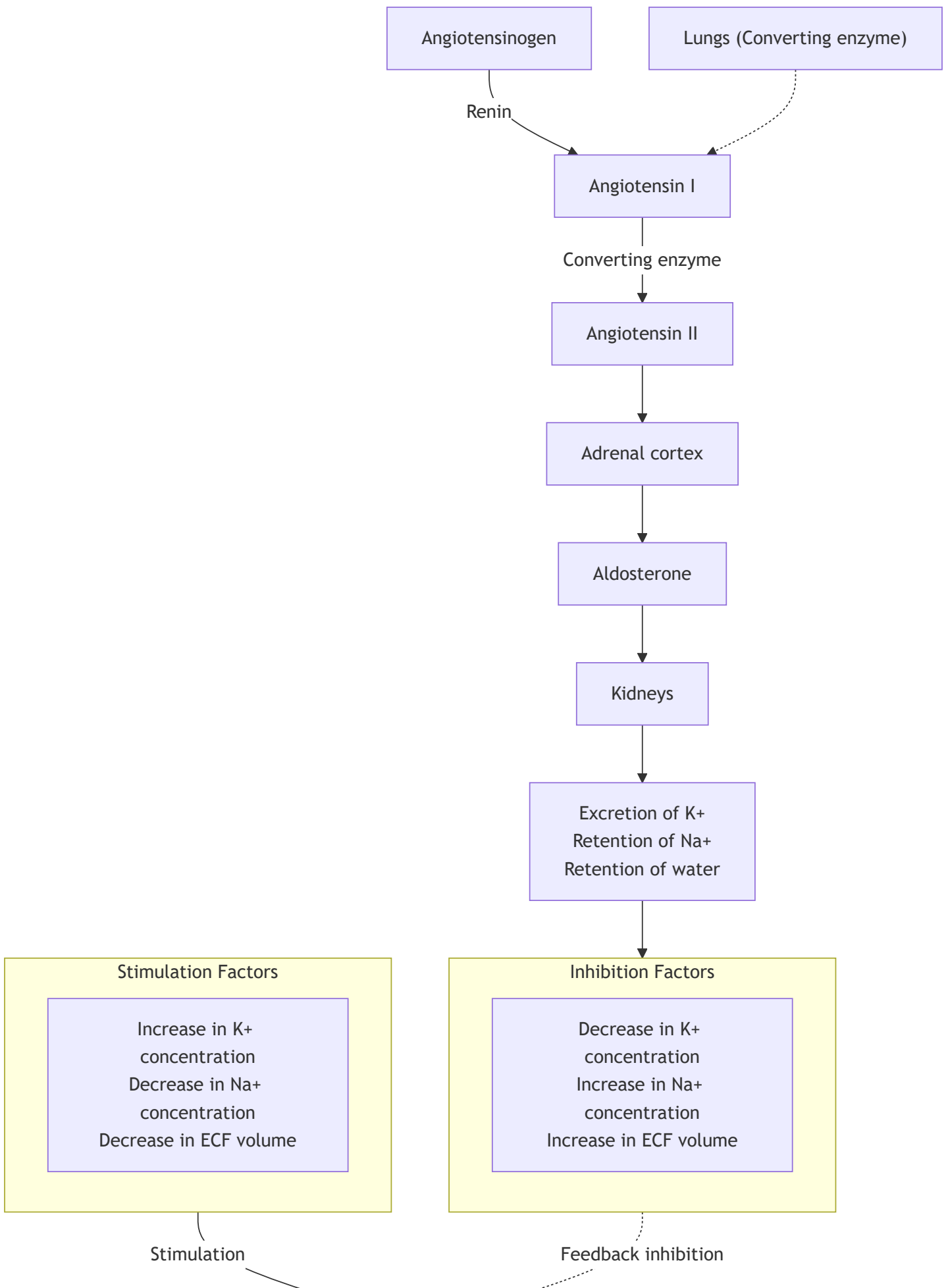


FIGURE 70.4: Importance of aldosterone. ECF = Extracellular fluid.

REGULATION OF SECRETION

- **Regulated by 4 important factors :—**
 - a. ↑se in K^+ concentration in ECF
 - b. ↓se in Na^+ concentration in ECF
 - c. ↓se in ECF volume
 - d. Adrenocorticotrophic hormone (ACTH)



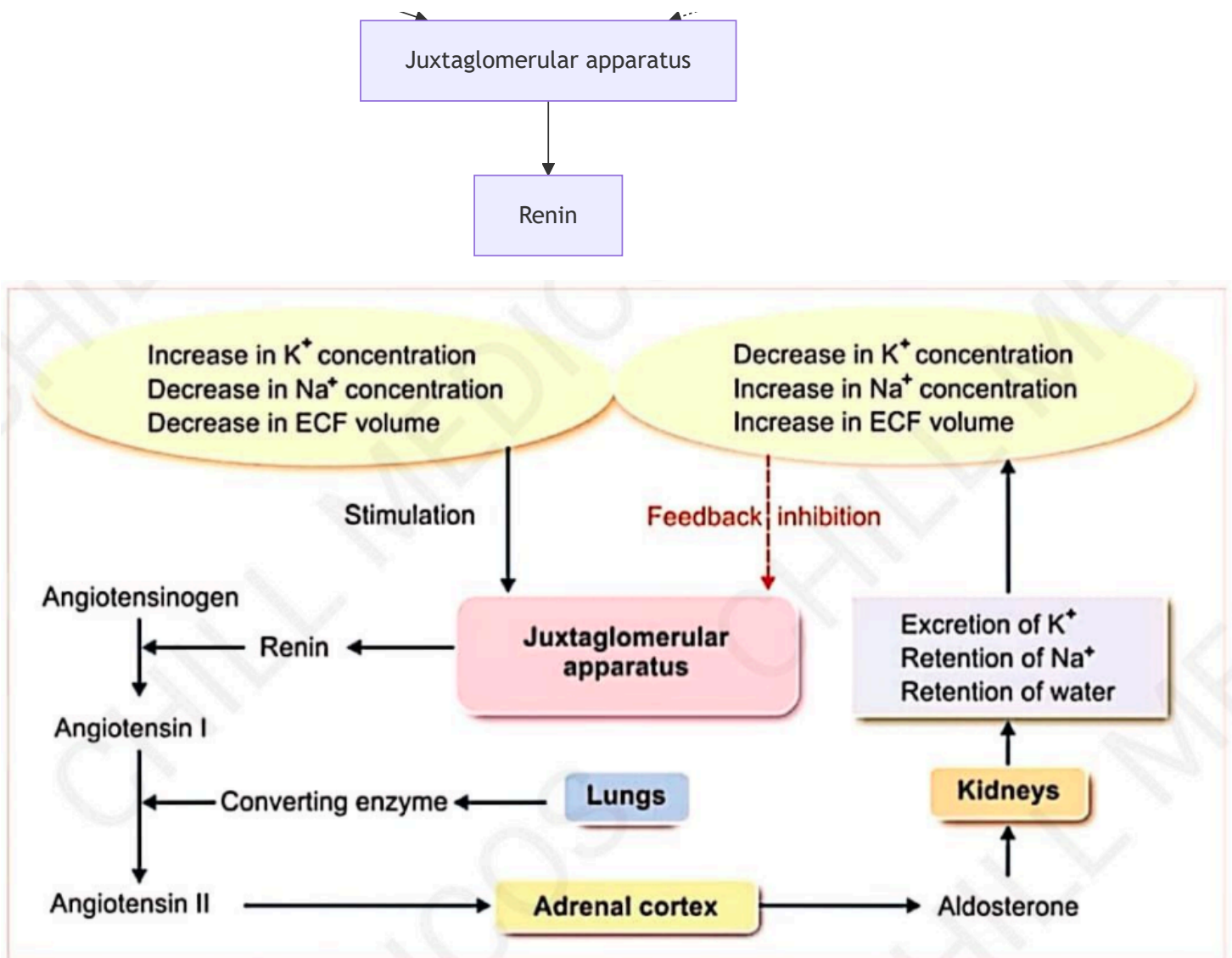


FIGURE 70.5: Regulation of aldosterone secretion